

Flyhomes BBYS Calculator

Calculation Logic — Engineering Handoff Spec

Source of truth: `index.html` (single-file prototype), as implemented on 2026-08-13 · Verified against live JS, not just design docs · Supersedes the 2026-08-11 revision

Purpose. This document specifies the calculation and decision logic exactly as implemented in the working prototype (`index.html`), so it can be re-implemented in production code. Where the prototype's behavior differs from — or isn't covered by — the design doc (`CLAUDE.md`), that is called out explicitly with a **GAP** or **BUG** tag. Items resolved since the last revision are tagged **RESOLVED** . Everything else is marked **MATCHES DOC** .

Status as of this revision: all 5 issues raised in the 2026-08-11 spec have been resolved (product decisions confirmed, code fixed, or verified in-browser) — see §6 for the full change log. One additional edge case was found and fixed during this pass (§6.6). No open gaps or bugs as of 2026-08-13.

Contents

- 1. State model
- 2. Challenge selection → product rule engine
- 3. Core product formulas (Instant Equity, Cash Offer, Cross Collateral)
- 4. Fee formulas (origination, broker, GBC, accrued interest)
- 5. Estimated Savings formulas
- 6. Change log: issues found & resolved (2026-08-11 → 2026-08-13)
- 7. Appendix: constants reference

1. State model

A single mutable state object `S` drives every calculation. Relevant fields ([index.html:1150–1178](index.html:1150-1178)):

Field	Meaning	Default
<code>homeValue</code>	Departing home value (current home) — also the GBC fee-tier input	\$1,200,000
<code>mortgage</code>	Current mortgage balance on departing home	\$300,000
<code>newPrice</code>	Estimated purchase price of new home	\$800,000
<code>assets</code>	Liquid cash assets available	\$40,000
<code>transitionDays</code>	Days between buying new home and selling old one. User-editable, min 1	60
<code>gbc</code> , <code>cash</code> , <code>noCash</code> , <code>downsize</code>	Resolved product flags for the current render (derived — see §2)	—
<code>cashIntent</code> , <code>gbc_base</code> , <code>noCash_base</code>	Raw user intent flags, before Rule 3/5 gating resolves them	—
<code>forceCashOffer</code>	"Option A" manual override: user chose standalone Cash Offer over an auto-recommended Cross Collateral. Reset to <code>false</code> whenever challenge selection changes	false
<code>naturalIsCC</code>	True if Cross Collateral would be recommended <i>without</i> the override — drives the switch-offer banner	—
<code>statesDiffer</code>	True when "current home state" and "buying state" dropdowns are both filled and unequal	false
<code>flyhomesLoan</code>	True = GBC bundle rate; false = GBC standalone rate	true
<code>savCashPct</code> / <code>savStagedPct</code> / <code>savHpaPct</code>	Savings-section sliders	3.5% / 5% / 4%
<code>savStagingCost</code>	Flat staging cost deducted from staged-home premium	\$3,500
<code>savMovingFee</code> , <code>savHousing</code> , <code>savStorage</code>	"Move Once" line items — auto-computed, user-editable	derived from <code>homeValue</code>

No state fields were added or removed since the 2026-08-11 revision. The dead `savings` field previously returned from `calc()` (not part of `S`) has been deleted — see §6.4.

1.1 Home-value input ceiling (new since 2026-08-11)

"Estimated Value of Your Current Home" is capped at **\$2,000,000** — both the slider max (<index.html:325>, was \$3M) and the effective usable range of the paired free-text field. The text field itself has no hard-coded max (`parseVal()` , <index.html:1142>, accepts any typed number), so `render()` enforces the ceiling in code instead:

```
function render() {
  const overHomeValueLimit = S.homeValue > 2_000_000;
  $('homeValue-limit-warning').classList.toggle('hidden', !overHomeValueLimit);
  $('results-panel').classList.toggle('hidden', overHomeValueLimit);
  if (overHomeValueLimit) {
    $('cash-warning').classList.add('hidden'); // clear stale banner from Step 1
    return; // skip calc()/render entirely
  }
  ...
}
```

index.html:1544–1558. When `S.homeValue > 2,000,000`, the entire results panel (`#results-panel`) is hidden, a "Please contact us to learn more." message (`#homeValue-limit-warning`) is shown under the input, and `calc()` / `applyRule3Gating()` are never invoked for that render pass — no estimate is attempted above this threshold.

Product decision (confirmed 2026-08-13): departing-home values above \$2,000,000 fall outside the GBC Fee Matrix's standard tiers (CLAUDE.md §5C) and require manual/LO review. Rather than surfacing a partial or exception-riddled estimate, the tool declines to estimate at all above this threshold and directs the user to contact Flyhomes directly. This also makes the GBC >\$2M "exception" code path (§4.3) practically unreachable through normal use — see §6.5.

2. Challenge selection → product rule engine

Four challenge checkboxes map to internal ids: `dti`, `equity`, `cash`, `downsize` ([index.html:1654–1660](#)), stored in a Set called `selected`. Two functions resolve them into product flags on every change:

`syncStateFromChallenges()` then `applyRule3Gating()`. `applyRule3Gating()` also re-runs on every `render()` so slider changes (price, mortgage, home value) can flip the recommendation live.

2.1 `syncStateFromChallenges()` [index.html:1662–1676](#)

```
isDownsize      = selected.has('downsize')
isCrossCollateral = (selected.has('equity') && selected.has('cash')) || isDownsize

S.downsize      = isDownsize
S.noCash_base   = isCrossCollateral
S.cashIntent    = selected.has('cash') && !isCrossCollateral
S.gbc_base      = selected.has('dti') && !isCrossCollateral
S.forceCashOffer = false    // reset override on any challenge-card change
```

Matches design doc Rule 0 (downsize overrides) and Rule 1 (equity+cash → Cross Collateral).

[MATCHES DOC](#)

2.2 `applyRule3Gating()` [index.html:1681–1716](#)

```
isRule3CC = false
if (cashIntent && newPrice > 0):
    projectedFinalLTV = (newPrice + mortgage - homeValue * 0.9) / newPrice
    isRule3CC = projectedFinalLTV <= 0.75

if (statesDiffer):
    // Rule 5 – cross-state block
    if (noCash_base):
        // a same-state CC would have won
        if (!downsize): cashIntent = true // Rule 1 combo → keep the cash-offer intent
        gbc_base = true                  // both fallbacks pair with DTI Buster
    noCash_base = false
    isRule3CC = false

naturalIsCC = noCash_base || isRule3CC

if (naturalIsCC && !downsize && forceCashOffer):
    // Option A: user overrode CC → standalone Cash Offer
    noCash = false; cash = true; gbc = gbc_base
else:
    noCash = naturalIsCC
    cash = cashIntent && !isRule3CC
    gbc = isRule3CC ? false : gbc_base    // CC can't pair with GBC
```

Matches design doc Rule 3 (auto-gate on `projectedFinalLTV`, with the "Keep More Cash" override) and Rule 5 (cross-state block). [MATCHES DOC](#)

RESOLVED

since 2026-08-11: the cross-state block now distinguishes *why* Cross Collateral would have won before falling back — a pure Rule 0 downsize intent falls back to **Instant Equity + DTI Buster** (downsize alone carries no competitive-cash-offer intent), while a Rule 1 equity+cash combo falls back to **Cash Offer + DTI Buster**

(the user explicitly asked for a competitive cash offer). Previously, cross-state blocked *any* would-be-CC scenario by zeroing `noCash_base` without restoring *any* fallback intent for the pure-downsize case — silently dropping the user into Instant Equity with no GBC pairing and, per §2.3, a mismatched label. Full write-up in §6.1.

2.3 Card visibility & labels — `applyProductLabel()` [index.html:1718–1755](#)

Condition	Result
<code>dtiOnly = selected.size===1 && has('dti')</code>	Hides main solution card, cost/benefit card, CTA card, number card. Shows only the GBC/DTI Buster card.
<code>noneSelected = selected.size===0</code>	Same as above — nothing shown.
<code>showGBC = !noneSelected && !isCrossCollat</code>	GBC (DTI Buster) card shows whenever anything is selected and the resolved product isn't Cross Collateral — regardless of whether "dti" was checked. This is what implements Rule 2/Rule 3's "+ DTI Buster" pairing.
<code>mostLikely = showGBC && !has('dti')</code>	"Most likely needed" badge on the DTI Buster card — shown when GBC is auto-added by Rule 2/3/5 rather than manually checked.

RESOLVED since 2026-08-11: `applyProductLabel()` no longer independently derives `solutionLabel` / `productLabel` / `intro` from raw checkbox state. The solution card's title, tooltip text, and "same-state required" note are now set exactly once, in `render()` ([index.html:1557–1560](#)), directly from `calc()`'s resolved `r.solutionName` / `r.productName` / `r.productIntro` — the same branch (§3) that computes every dollar figure on the card. There is now a single source of truth for "what product is this," eliminating the class of bug where the label and the underlying math could disagree (§6.1). `applyProductLabel()`'s remaining responsibility is card visibility and the GBC badge only.

3. Core product formulas

Computed in `calc()` [index.html:1197–1263](#), branching on the resolved `S.noCash` / `S.cash` flags from §2:

```
availableEquity  = max(0, homeValue * 0.75 * 0.9 - mortgage)  // = Instant Equity value
combinedResources = assets + availableEquity
cashThreshold    = newPrice * 0.05
cashEligible     = combinedResources >= cashThreshold
gbcActive        = S.gbc && !noCash  // GBC never pairs with Cross Collateral
```

3.1 Branch A — Cross Collateral (if `noCash` is true)

```
ccCltv  = (newPrice + homeValue) * 0.75 - mortgage
maxLoan = max(0, min(newPrice * 1.05, ccCltv))
ltvAcq  = newPrice > 0 ? maxLoan / newPrice : 0
originationRate = ltvAcq > 0.90 ? 0.015 : 0.010
brokerFee      = maxLoan * 0.01

// Label:
if (S.downsize): solutionName = "Retire & Downsize"      productName = "Cross Collateral"
else:            solutionName = "Move with $0 out of pocket" productName = "Cross Collateral"
```

Matches design doc §5E Cross Collateral formula and §5B origination-fee tier. [MATCHES DOC](#)

3.2 Branch B — Cash Offer (`else if cash && cashEligible`)

```
maxPurchasePrice = combinedResources / 0.05
maxLoan          = max(0, min(newPrice, maxPurchasePrice) * 0.95)
ltv              = newPrice > 0 ? maxLoan / newPrice : 0
originationRate  = ltv > 0.90 ? 0.015 : 0.010
brokerFee        = maxLoan * 0.01
solutionName     = "All-Cash Advantage"  productName = "Cash Offer"
```

Matches design doc §5E "BBYS + Cash Offer" and Max Purchase Price formulas. [MATCHES DOC](#)

3.3 Branch C — Instant Equity (else — default / fallback)

```
maxLoan  = max(0, homeValue * 0.75 * 0.9 - mortgage)
originationRate = 0.02  // 1st lien rate, applied unconditionally – see §6.2
brokerFee      = maxLoan * 0.005
solutionName   = "Equity for Down Payment"  productName = "Instant Equity"
```

Matches design doc §5E Instant Equity formula and §5B (1st-lien rate). [MATCHES DOC](#)

RESOLVED (confirmed, not implemented)

since 2026-08-11: product confirmed this calculator is a

directional/preliminary estimate, not a bindable quote — lien-position detection is intentionally out of scope. There is no reliable signal to derive it from: `Current Mortgage Balance > 0` does not necessarily mean the Instant Equity

loan lands in 2nd lien, since many equity-advance products require 1st-lien position and would pay off/subordinate the existing mortgage as part of funding — a lending-structure decision this app doesn't (and won't) model. `originationRate` stays hardcoded to `0.02` for every Instant Equity scenario. See CLAUDE.md §5B for the recorded decision. Full write-up in §6.2.

This branch is also the fallback whenever `cash` is true but `cashEligible` is false (insufficient combined cash + equity for the 5% down payment) — in that case a "cash warning" banner is additionally shown ([index.html:1606–1612](#)) with the shortfall amount: `gap = cashThreshold - combinedResources`.

4. Fee formulas

4.1 Origination fee

```
originationFee = maxLoan * originationRate // rate resolved per branch above, §3
```

4.2 Broker fee

Rolled into "Other Costs" for display — never shown as a standalone line item to the buyer, per design doc §5D. Rate depends on product track (see §3): Cross Collateral 1% of capped loan, Cash Offer 1% of FCO loan amount, Instant Equity 0.5% of advance amount.

4.3 GBC fee — `getGbcFee(homeValue, isBundle)` <index.html:1183-1196>

```
if      (homeValue <= 500,000):  standalone = 2,500;  credit = 0
elif    (homeValue <= 750,000):  standalone = 3,500;  credit = 1,000
elif    (homeValue <= 1,000,000): standalone = 5,000;  credit = 2,500
elif    (homeValue <= 1,500,000): standalone = 7,500;  credit = 2,500
elif    (homeValue <= 2,000,000): standalone = 10,000; credit = 2,500
else:                                     return null // >$2M exception – manual review required

return isBundle ? (standalone - credit) : standalone
```

`isBundle` passed in as `!dtiOnly && S.flyhomesLoan` — i.e. the bundle (net) rate only applies when a Flyhomes loan product is shown alongside GBC and it isn't the DTI-only scenario. In DTI-only mode, GBC is always priced standalone (no loan bundle exists to net against). **MATCHES DOC** §5C exactly, including the >\$2M exception returning `null`.

Practical note: since §1.1's home-value input ceiling now blocks any render above `homeValue > 2,000,000` before `calc()` is ever called, this `else: return null` branch is effectively unreachable through the current UI. It remains in code as a defensive fallback — worth knowing about if a future change (e.g. a different field feeding `getGbcFee`, or the ceiling being relaxed) reopens this path.

4.4 Accrued interest

```
accruedInterest = maxLoan * 0.0999 * (transitionDays / 365)
```

Only shown/rendered when `maxLoan > 0` (hidden for DTI-only). **MATCHES DOC**

4.5 Total estimated cost

```
gbcFeeNum = (gbcFee === null) ? 0 : (gbcFee ?? 0) // >$2M exception treated as $0 in the total
total      = originationFee + brokerFee + gbcFeeNum + accruedInterest
```


RESOLVED (differently)

since 2026-08-11: the total still technically treats an unresolved GBC exception as \$0 rather than flagging as incomplete — but per §1.1, the UI ceiling now prevents that state from ever being reached in practice, since `homeValue` can no longer exceed \$2,000,000 through the input. Product chose to close this gap upstream (block the input) rather than downstream (flag an incomplete total). Full write-up in §6.5.

5. Estimated Savings formulas

Computed in `calcSavings()` `index.html:1325–1337`:

5.1 Cash Offer Price Advantage

```
isCash      = cash || noCash
cashDiscount = isCash ? newPrice * (savCashPct / 100) : 0
```

Matches design doc §5F.1: shows whenever the recommended product uses a cash offer — Cash Offer track (`S.cash`) or Cross Collateral / Retire & Downsize (`S.noCash`). **MATCHES DOC**

RESOLVED since 2026-08-11: previously the eligibility check additionally required `assets >= newPrice * 0.05` — i.e. liquid cash *alone* had to cover the 5% down payment, not the combined cash+equity figure (`combinedResources`) used everywhere else in the calculator for cash eligibility. That extra gate was removed after confirming with product: any user recommended a cash-offer-based product should see this benefit, regardless of whether the cash came from liquid assets or unlocked equity. Full write-up in §6.3.

5.2 Staged Home Premium

```
staged = max(0, homeValue * (savStagedPct / 100) - savStagingCost)
```

Matches design doc tooltip (net of \$3,500 staging cost) — note the doc's "Formula" line omits the staging-cost subtraction and only mentions it in the tooltip text; the code implements the tooltip version. **MATCHES DOC** (tooltip version)

5.3 Home Price Appreciation (HPA)

```
hpa = newPrice * (savHpaPct / 100) * (transitionDays / 30) / 12
```

Algebraically identical to design doc's `newHomePrice * (hpaRate% / 12) * (transitionDays / 30)` — order of operations differs, result is the same. Buy-side only, per doc. **MATCHES DOC**

5.4 "Move Once, Not Twice" (logistics)

```
// Auto-computed defaults (index.html:1398–1400), independently user-editable after:
savMovingFee = round(homeValue * 0.003 + 500)
savHousing   = round(homeValue * 0.006 * (transitionDays / 30))
savStorage   = round((150 + homeValue * 0.0002) * (transitionDays / 30))
logistics    = savMovingFee + savHousing + savStorage // uses live (possibly edited) field values
```

MATCHES DOC exactly.

5.5 Grand total

```
total = cashDiscount + staged + hpa + logistics
```

The DTI-only ("GBC-only") card computes its own total that always excludes the cash discount row (since Cash Offer is never active in DTI-only mode): `g-total = staged + hpa + logistics` .

The unused `savings = max(0, maxLoan * 0.04 - total)` field previously returned from `calc()` (a separate, unrelated "4% return" figure — not part of this section) has been deleted; see §6.4.

6. Change log: issues found & resolved (2026-08-11 → 2026-08-13)

Every issue raised in the 2026-08-11 revision of this spec was reviewed with product and closed out. One additional edge case surfaced during this pass's re-audit and was fixed in the same session. **No open gaps or bugs remain as of this revision.**

RESOLVED 6.1 Downsize + cross-state produced mismatched label vs. math

Design doc Rule 0 states downsize "overrides all other logic combinations." The implementation had Rule 5's cross-state block run after Rule 0 and unconditionally zero `noCash_base` — including when `S.downsize` was true — while `applyProductLabel()` checked `S.downsize` first and unconditionally, so the card kept showing "Retire & Downsize" / "Cross Collateral" even though every dollar figure had silently fallen through to Instant Equity math.

Fix: confirmed with product that Cross Collateral requires same-state homes with no exception for downsize (CLAUDE.md Rule 5 now states this explicitly). `applyRule3Gating()` was updated so a blocked downsize falls back to **Instant Equity + DTI Buster** (matching the "use equity" intent, since downsize alone carries no competitive-cash-offer intent — Cash Offer would have solved the wrong problem), while a blocked Rule 1 equity+cash combo still falls back to **Cash Offer + DTI Buster** as before. The duplicate, out-of-sync label logic in `applyProductLabel()` was deleted entirely — labels now come from the same resolved `calc()` branch that computes the numbers (§2.3), so this class of bug can't recur.

RESOLVED (confirmed, not implemented) 6.2 Instant Equity 2nd-lien rate

Design doc §5B specifies 2.0% for 1st lien and 2.5% for 2nd lien. Confirmed with product this is an intentional simplification, not an oversight — see §3.3. This is a preliminary/directional estimate tool, not a bindable quote, and there's no reliable way to infer lien position from the existing inputs (a non-zero `Current Mortgage Balance` does not reliably imply 2nd lien — see CLAUDE.md §5B for the reasoning). `originationRate` stays hardcoded at 2.0% for all Instant Equity scenarios.

RESOLVED 6.3 Cash-discount eligibility differed between main card and savings section

Same root cause and same fix as §6.1's savings-section counterpart, §5.1: the savings section's cash-discount visibility used raw `assets` instead of `combinedResources`. Fixed alongside §5.1.

RESOLVED 6.4 Dead code: unused savings field in calc()

`const savings = max(0, maxLoan * 0.04 - total)` was returned from `calc()` but never read or rendered anywhere in the app — a leftover from an earlier design iteration. Confirmed nothing depends on it and deleted, along with the now-unused `assets` destructure in `calcSavings()`.

RESOLVED (differently) 6.5 GBC fee >\$2M exception silently excluded from grand total

The GBC line item correctly showed "Exception — contact us" above \$2M, but the overall Total Estimated Cost silently treated the unresolved exception as \$0 rather than flagging the total as incomplete. Rather than teaching the total-cost calculation to surface a partial/incomplete state, product chose to block the input itself: see §1.1's new \$2,000,000 ceiling on "Estimated Value of Your Current Home." Above that value, no estimate is attempted at all — the entire results panel is hidden and the user is directed to contact Flyhomes. This closes the gap upstream of where it was originally found.

6.6 (new) Stale cash-warning banner when crossing the \$2M ceiling

Found during this pass's re-audit while implementing §1.1: the "cash offer needs at least 5% in liquid assets" warning banner (`#cash-warning`) lives in the Step 1 card, outside `#results-panel` . If it was already showing and the user then pushed "Estimated Value of Your Current Home" above \$2,000,000, the early return in `render()` skipped the block that updates it ([index.html:1606–1612](#)), leaving a stale "you need \$X more in cash or home equity" message displayed alongside the new "Please contact us" notice. **Fix:** the \$2M-ceiling branch in `render()` now explicitly hides `#cash-warning` before returning ([index.html:1548–1554](#)).

7. Appendix: constants reference

Constant	Value	Used in
Accrued interest rate	9.99% / yr	Total cost — all tracks
Instant Equity multiplier	$0.75 \times 0.9 = 0.675$	Instant Equity value / available equity
Cash Offer max LTV	95%	Cash Offer max loan cap
Cash Offer min down payment	5%	Cash eligibility threshold
Cross Collateral acquiring LTV cap	105%	Cross Collateral max loan cap
Cross Collateral combined LTV	75%	Cross Collateral CLTV formula
Origination fee — CO / CC, LTV > 90%	1.5%	Origination fee
Origination fee — CO / CC, LTV ≤ 90%	1.0%	Origination fee
Origination fee — Instant Equity	2.0% (all scenarios)	Origination fee — 2nd-lien rate intentionally not implemented, confirmed §6.2
Broker fee — Cross Collateral	1% of capped loan	Rolled into Other Costs
Broker fee — Cash Offer	1% of FCO loan amount	Rolled into Other Costs
Broker fee — Instant Equity	0.5% of advance amount	Rolled into Other Costs
Rule 3 gating threshold	$\text{projectedFinalLTV} \leq 0.75 \rightarrow \text{Cross Collateral}$	Auto-recommendation between CC and CO
Rule 5 cross-state fallback	Downsize → Instant Equity+GBC; equity+cash combo → Cash Offer+GBC	Cross-state block fallback (§2.2, §6.1)
Home-value input ceiling	\$2,000,000	Blocks results panel + GBC > \$2M exception path (§1.1, §6.5)
Staging cost (flat, editable)	\$3,500	Staged Home Premium
Default sliders	Cash discount 3.5%, Staged premium 5%, HPA 4%	Estimated Savings
Default transition days	60 (min 1)	Accrued interest, HPA, temp housing, storage

